Project Title

Smart Student Management Web Application

Project Overview (Minimal)

A responsive web application to manage student registration, marks, profile, and reports using HTML5, CSS3, JavaScript, jQuery, AJAX, Bootstrap, with LocalStorage, IndexedDB, JSON, and DOM manipulation.

Technology Stack

- **HTML5** – Structure
- **CSS3 + Animations + Tooltips** – Styling
- **Bootstrap** – Responsive UI
- **JavaScript (ES6)** – Logic
- **jQuery** – DOM & Events
- **AJAX** – Server communication
- **JSON** – Data exchange
- **LocalStorage & IndexedDB** – Client storage

Application Modules & Topic Mapping

1.Student Registration Module

Features

- Student Name, Age, Email, Course, Marks
- Validation & Events
- Save data locally

2.Student Dashboard

Features

- Display students in table
- Sort by marks
- Filter pass/fail

- Highlight low scores

3.Marks Analysis Module

Features

- Total, Average, Grade
- Random test data
- Date-wise entries

4.Persistent Storage Module

Features

- Save large data offline
- Retrieve on reload

5. Edit & Delete Students

Features

- Edit record
- Delete record
- Confirm popup

6. Advanced JavaScript Logic

Features

- Inheritance-based Student model
- Reusable functions

7. Styling & UI Enhancements

Features

- Hover effects
- Animated buttons
- Responsive layout

8. AJAX Data Sync Module

Features

- Load student data from server
- Save using backend API

9. Browser Interaction Module

Features

- Detect browser
- Screen size
- Navigation history

10. **Validation & Error Handling**

Features

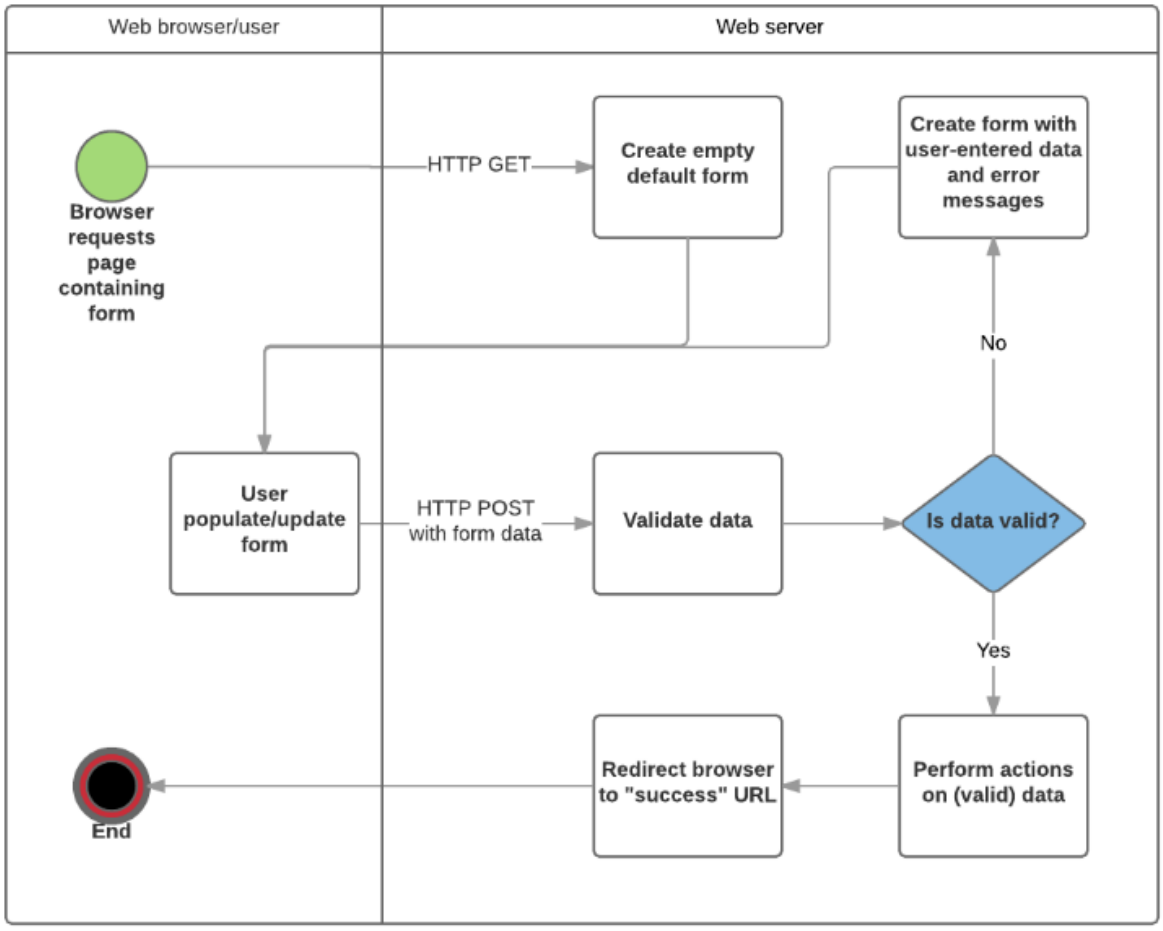
- Form errors
- Debug messages

Folder Structure

```
/project
  ├── index.html
  ├── css/
  │   └── style.css
  ├── js/
  │   ├── app.js
  │   ├── storage.js
  │   ├── ajax.js
  ├── data/
  │   └── students.json
  └── server/
```

└── api.php / api.asp

Student Registration Module – Flowchart



FLOW:

Student Object

↓

JSON.stringify()

↓

LocalStorage (small data)

↓

IndexedDB (large data)

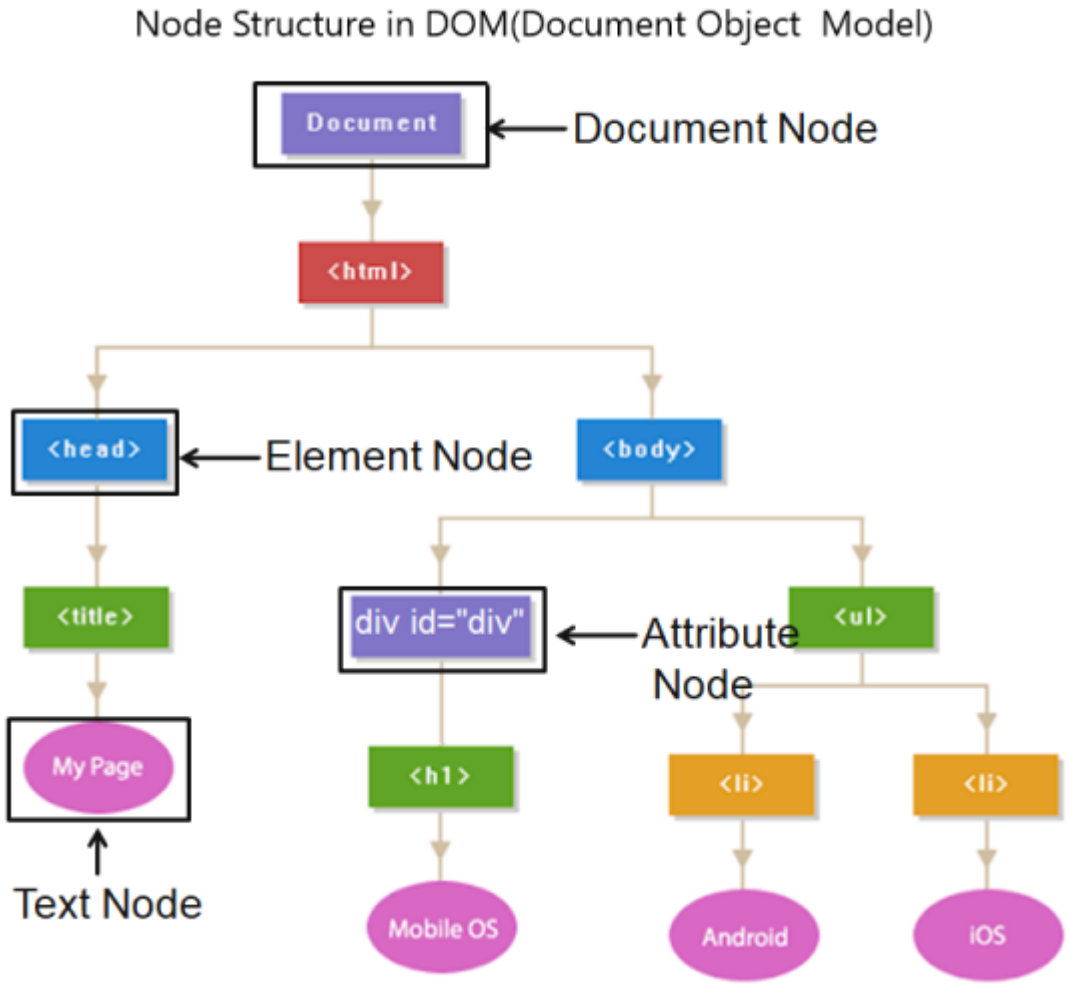
↓

Retrieve on Page Load

↓

JSON.parse()

DOM & jQuery Interaction Flow



FLOW:
User Action

↓

XMLHttpRequest

↓

Send JSON Data

↓

Server (PHP / ASP)

↓

Database

↓

JSON Response

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Update UI

Overall Module Interaction Diagram

UI (HTML + CSS + Bootstrap)

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JavaScript Logic

↓

DOM / jQuery

↓

LocalStorage / IndexedDB

↓

AJAX

↓

Server API

↓

Database